

Tech Stack Definition: AgriTech Marketplace MVP

1. Frontend (Web App transitioning to PWA)

Owner: Frederick Abila

- **Core Framework:** React.js powered by Vite.
 - *Why?* Vite is incredibly fast for development. More importantly, using the vite-plugin-pwa makes converting this React app into a fully functional PWA (with service workers for offline caching) literally a 5-minute task after you get shortlisted.
- **Styling:** Tailwind CSS.
 - *Why?* Pre-built utility classes mean Frederick can build a beautiful, responsive mobile-first UI for farmers and drivers without writing custom CSS from scratch.
- **Maps/Routing:** Mapbox GL JS.
 - *Why?* Highly customizable, great free tier, and perfect for the logistics tracking and driver routing workflows.
- **State Management:** Zustand or React Context (Keep it lightweight, avoid Redux for the MVP).

2. Backend & API (Core Infrastructure)

Owner: Daniel Darko Danquah & Nana Kojo (AI)

- **Core Framework:** Python with FastAPI.
 - *Why?* This is the secret weapon for your team. Daniel can build blazing-fast REST APIs, and because it's Python, Kojo's XGBoost machine learning models can be integrated *directly* into the backend without needing a complicated microservices architecture. FastAPI also auto-generates API documentation (Swagger), making frontend integration easy for Frederick.
- **Real-time Communication:** WebSockets (built into FastAPI).
 - *Why?* Necessary for the logistics module so the buyer sees the driver's location update in real-time without refreshing the page.

3. Database & Authentication

Owner: Daniel Darko Danquah

- **Platform:** Supabase (Backend-as-a-Service running PostgreSQL).
 - *Why?* Supabase is an open-source Firebase alternative but uses a relational SQL database.
 - **Crucial Feature:** It comes with **PostGIS** pre-installed. This allows you to do complex geographic queries instantly (e.g., "Find all available drivers within a 10km radius of the farmer in Tarkwa"). It also provides instant user authentication and real-time database

subscriptions out of the box, saving Daniel days of setup time.

4. AI & Machine Learning Module (Bonus Features)

Owner: Nana Kojo

- **Libraries:** Scikit-learn, XGBoost, Pandas.
- **Integration:** Export the trained XGBoost model as a .pkl or .json file and load it directly into Daniel's FastAPI backend to serve the `predict_transport_cost()` endpoint.

5. External APIs (Ghana Context)

- **Payments (MoMo):** Paystack API. (Easiest developer experience for integrating MTN/Telecel/AT Mobile Money).
- **Offline Access (USSD/SMS):** Africa's Talking API. (The industry standard for building USSD menus for farmers without internet).